

Project Design Phase

Solution Architecture

Date	19 Jul 2026
Team ID	
Project Name	AI-powered Job Recruitment & Candidate Management System
Maximum Marks	4 Marks

Solution Architecture

The **AI-Powered Job Recruitment & Candidate Management System** is developed on the **Salesforce Cloud Platform** to automate the complete recruitment lifecycle. The architecture integrates Salesforce CRM, Agentforce AI, Prompt Builder, Flow Builder, Approval Processes, and Lightning Experience to provide an intelligent and scalable recruitment solution.

The system consists of multiple layers:

1. Presentation Layer

- Salesforce Lightning Experience
- Web Browser
- Mobile Browser

Users:

- Candidate
- Recruiter
- Hiring Manager
- Administrator

2. Application Layer

This layer manages all business logic.

Functions include:

- User Authentication
- Candidate Registration
- Job Opening Management
- Resume Management

- Candidate Application
- Interview Scheduling
- Approval Process
- Record Triggered Flows
- Screen Flows
- Validation Rules

3. AI Layer

The intelligent layer provides AI-powered recruitment assistance.

Components:

- Agentforce AI
- Prompt Builder
- Candidate Summary Generation
- Hiring Risk Analysis
- AI Recommendation Engine

4. Data Layer

Salesforce CRM stores all project information.

Objects include:

- Candidate
- Candidate Application
- Job Opening
- Interview
- Recruiter
- Hiring Manager
- Reports
- Dashboards

5. Integration Layer

Provides communication with external services.

Services include:

- Salesforce Email Service
- Salesforce Identity
- Notification Service
- Salesforce Files
- Reports & Dashboards

6. Security Layer

Security is implemented using Salesforce features.

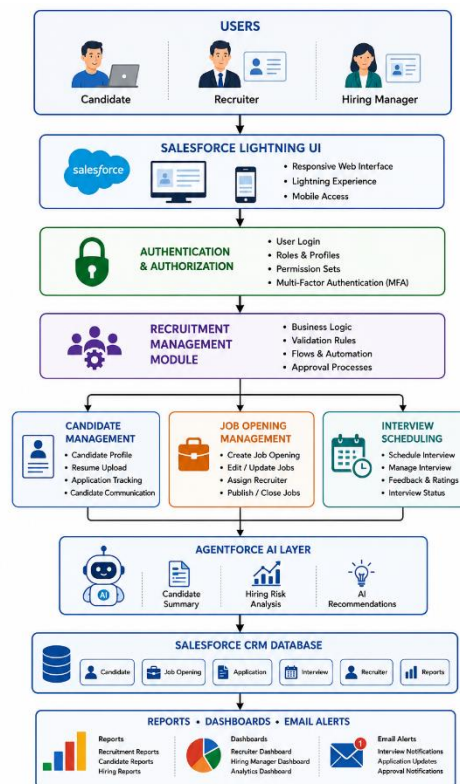
- Role Hierarchy
- Profiles
- Permission Sets
- Multi-Factor Authentication (MFA)
- Field Level Security
- Record Level Security
- Encryption

7. Cloud Infrastructure

The application is deployed on the Salesforce Cloud Platform providing:

- High Availability
- Scalability
- Backup & Recovery
- Disaster Recovery
- Multi-Tenant Architecture

Architecture Workflow



Benefits of the Architecture

- Centralized recruitment management
- AI-assisted hiring decisions
- Automated recruitment workflow
- Secure cloud-based platform
- Real-time reports and dashboards
- High scalability
- Improved recruiter productivity
- Better candidate experience
- Faster hiring process
- Reduced manual effort

Technologies Used

Layer	Technology
Presentation	Salesforce Lightning Experience, HTML5, CSS3, JavaScript
Business Logic	Salesforce Flow Builder, Apex, Validation Rules
AI	Agentforce AI, Prompt Builder
Database	Salesforce CRM Objects
Storage	Salesforce Files
Notifications	Salesforce Email Services
Security	Profiles, Roles, Permission Sets, MFA
Cloud	Salesforce Cloud Platform

References

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